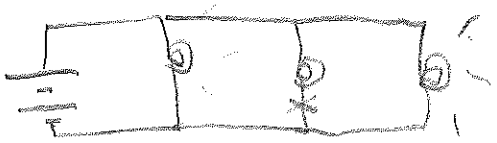
Practice Problems
Calculus Physics II

Chapter 43 Electric Circuits

Name: _____ **Date:** _____

1) 5 light bulbs are wired to a power supply, when one of the bulbs goes out, the other 4 bulbs stay on, what type of circuit are the bulbs wired?



A) series.
B) parallel.
C) either series or parallel.

2) Connect a pair of lamps in series and current is drawn from the battery. Connect the same lamps in parallel and the current drawn is

$$R_s = R_1 + R_2 + R_3 \cdots$$

$$R_p = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \right)}$$

A) less.
B) sometimes more, sometimes less.
C) more.
D) the same.

3) The equivalent resistance of any parallel branch in a circuit is

A) always less than the resistance of the lowest resistor.
B) often less than the resistance of the lowest resistor.
C) usually half the value of the lowest resistor.
D) none of the above

4) The junction rule, sum of the currents into a junction is equal to the sum of the currents out of the junction is a statement of

A) the law of conservation of momentum.
B) the law of conservation of charge.
C) the law of conservation of energy.
D) the law of conservation of angular momentum.
E) Newton's second law.

5) Four resistors of 20 Ω, 40 Ω, 60 Ω, and 80 Ω are connected in series across a 12 V dc power supply,

A) Determine the equivalent resistance of the circuit?

$$R_s = R_1 + R_2 + R_3 + R_4 = 200\,\Omega$$

B) Determine the current supplied by the dc power supply.

$$V = IR_s \qquad I = \frac{V}{R_s} = 0.06\,A$$

C) Determine the current through each of the 4 resistors.

$$I_1 = I_2 = I_3 = I_4 = 0.06\,A$$

D) Determine the voltage drop across each of the 4 resistors.

$V_1 = IR_1 = 1.2V$ | $V_3 = IR_3 = 3.6V$

$V_2 = IR_2 = 2.4A$ | $V_4 = IR_4 = 4.8V$

E) Determine the electrical power supplied to the circuit by the power supply.

$$P = IV = (0.064)(12V) = 0.72\,W$$

6) Four resistors of 20 Ω, 40 Ω, 60 Ω, and 80 Ω are connected in <u>parallel</u> across a 12 V dc power supply.

A) Determine the equivalent resistance of the circuit?

$$R_p = \frac{1}{\frac{1}{20\Omega} + \frac{1}{40\Omega} + \frac{1}{60\Omega} + \frac{1}{80\Omega}} = 9.6\Omega$$

B) Determine the current supplied by the dc power supply.

$$V = IR_p \qquad I = \frac{V}{R_p} = \frac{12V}{9.6\Omega} = 1.25A$$

C) Determine the current through each of the 4 resistors.

$I_1 = \frac{V}{R_1} = 0.6A$ | $I_3 = \frac{V}{R_3} = 0.2A$

$I_2 = \frac{V}{R_2} = 0.3A$ | $I_4 = \frac{V}{R_4} = 0.15A$

D) Determine the voltage drop across each of the 4 resistors.

$$V = 12V = V_1 = V_2 = V_3 = V_4$$

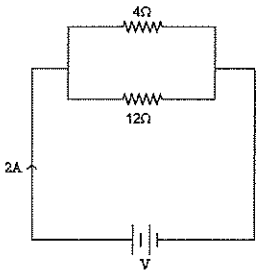
E) Determine the electrical power supplied to the circuit by the power supply.

$$P = IV = (1.25A)(12V) = 15\,W$$

7) A 4 Ω resistor is connected in parallel with a 12 Ω resistor and this combination is connected to a dc power supply with voltage V as shown in the figure. If the total current in this circuit is 2 A, what is the value of voltage V?

$$R_p = \frac{1}{\left(\frac{1}{4\Omega} + \frac{1}{12\Omega}\right)} = 3\Omega$$

$$V = IR_p = 6V$$



8) A 4 Ω resistor is connected in parallel with a 12 Ω resistor and this combination is connected to a dc power supply with voltage V as shown in the figure. If the total current in this circuit is 2 A, what is the current through the 12 Ω resistor?

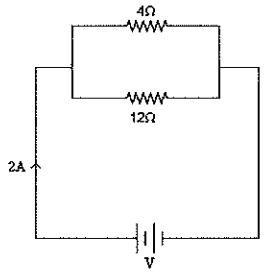
$V_{12\Omega} = 6V$

$R_{12\Omega} = 12\Omega$

$V_{12\Omega} = I_{12\Omega} R_{12\Omega}$

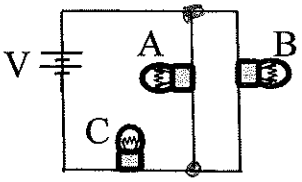
$I_{12\Omega} = \frac{V_{12\Omega}}{R_{12\Omega}} = \frac{6V}{12\Omega}$

$I_{12\Omega} = 0.5A$



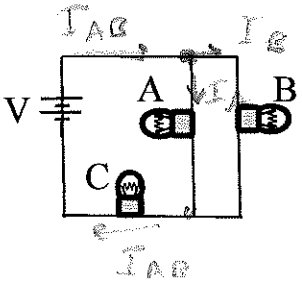
9) In the circuit shown, how many nodes are in the circuit?

A) 1
B) 2
C) 3
D) 4
E) 0



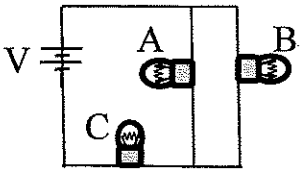
10) In the circuit shown, if all the light bulbs are identical, which bulb or bulbs have the largest current going through them?

A) Bulb A
B) Bulb B
C) Bulb C
D) Both Bulb A and B
E) They all have the same current through them.



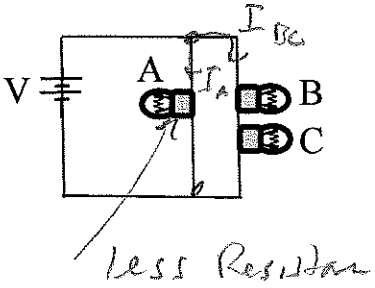
11) In the circuit shown, if all the light bulbs are identical, which bulb or bulbs will shine the brightest?

A) Bulb A
B) Bulb B
C) Bulb C
D) Both Bulb A and B
E) They will all shine at the same brightness.



12) In the circuit shown, if all the light bulbs are identical, which bulb will have the most current flowing through it?

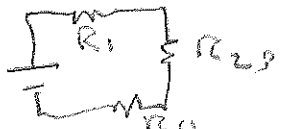
A) Bulb A
B) Bulb B
C) Bulb C
D) Both Bulb B and C
E) They will all have the same amount of current.

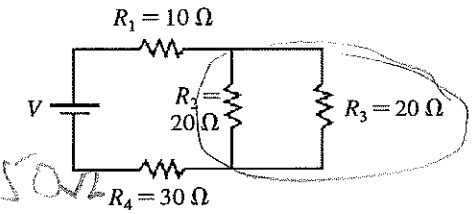


13) Determine the equivalent resistance in the circuit shown in the figure?

$R_{23} = \frac{1}{\frac{1}{R_2} + \frac{1}{R_3}} = 10\Omega$

$R_{1234} = R_1 + R_{23} + R_4 = 50\Omega$





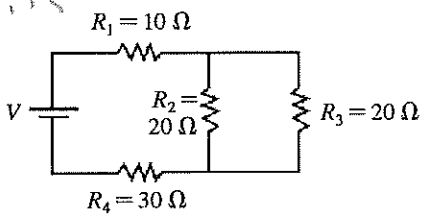
14) If $V = 40$ V and the battery is ideal, what is the current through R_1 in the figure?

$R_{eq} = 50\Omega$

$I = \frac{V}{R_{eq}} = 0.8A$

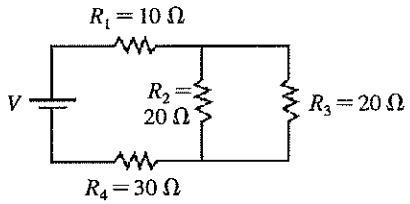
R_1 is in series with $R_{23} + R_4$

$I_1 = 0.8A$



15) If $V = 40$ V and the battery is ideal, what is the potential difference across R_1 in the figure?

$V_1 = IR_1 = (0.8A)(10\Omega) = 8V$

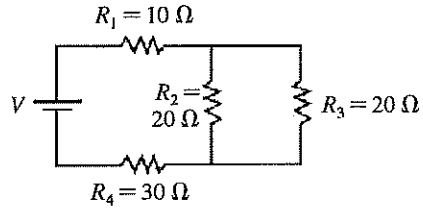


16) If $V = 40$ V and the battery is ideal, what is the potential difference across R_2 in the figure?

$R_{23} = 10\Omega$

$V_{23} = IR_{23} = 8V$

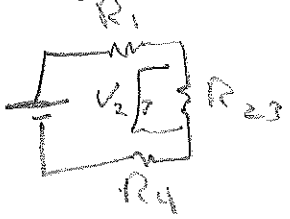




17) If $V = 40$ V and the battery is ideal, what is the current through R_3 in the figure?

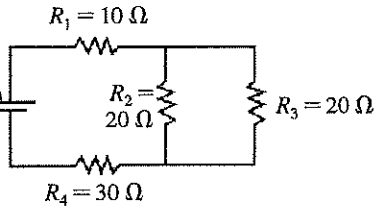
$V_{23} = 8V$

$R_3 = 20\Omega$



$V_{23} = I_3 R_3$

$I_3 = \frac{V_{23}}{R_3} = 0.4A$



18) Use Ohms Law and series/parallel combinations to solve for the current through each resistor R_1, R_2 and R_3.

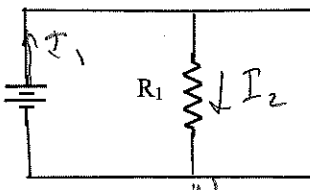
$R_{23} = R_2 + R_3 = 150\Omega$

$R_{123} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_{23}}\right)} = 37.5\Omega$

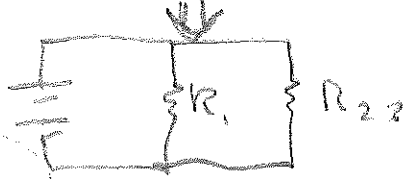
$I_1 = \frac{V}{R_{eq}} = 0.32A$

$I_2 = \frac{V}{R_1} = 0.24A$

$I_3 = \frac{V}{R_{23}} = 0.08A$

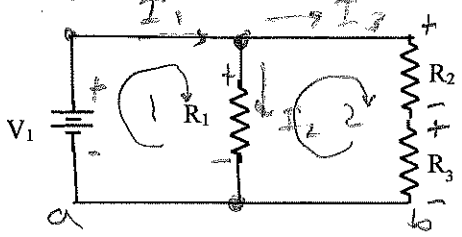


$V_1 = 12$ V
$R_1 = 50\ \Omega$
$R_2 = 100\ \Omega$
$R_3 = 50\ \Omega$



19) Use Kirchhoff's rules to solve for the current through each of the resistors R_1, R_2 and R_3.

$I_1 = I_2 + I_3 = 0.24A + 0.08A = 0.32A$



$V_1 = 12$ V
$R_1 = 50\ \Omega$
$R_2 = 100\ \Omega$
$R_3 = 50\ \Omega$

Loop #1 start at "a"

$(+V_1) + (-I_2R_1) = 0$

$I_2 = \frac{V_1}{R_1} = 0.24A$

Loop #2 start at "b"

$(+I_2R_1) + (-I_3R_2) + (-I_3R_3) = 0$

$I_3(R_2+R_3) = I_2R_1$

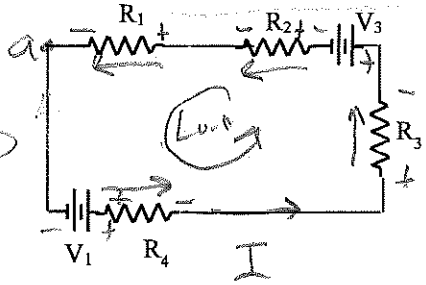
$I_3 = \frac{I_2R_1}{R_2+R_3} = 0.08A$

$I_3 = 0.08A$

20) Use Kirchhoff's rules to solve for the current in the circuit.

rise + drops = 0

$(+V_1) + (-IR_1) + (-IR_2) + (-V_3) \rightarrow +(-IR_2) + (-IR) = 0$



$V_1 = 10$ V
$V_3 = 5$ V
$R_1 = 100\ \Omega$
$R_2 = 200\ \Omega$
$R_3 = 50\ \Omega$
$R_4 = 200\ \Omega$

Solve for I

$I(R_1+R_2+R_3+R_4) = V_3 - V_1$

$I = \frac{V_3 - V_1}{(R_1+R_2+R_3+R_4)} = -0.009A$

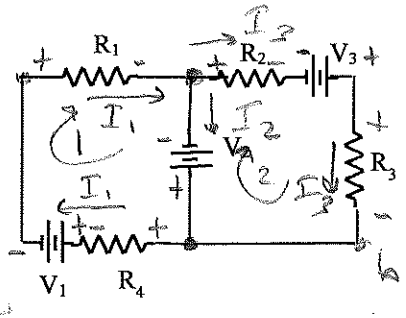
current opposite direction

21) Use Kirchhoff's rules to solve for the current through each of the resistors R_1, R_2 and V_2.

$I_1 = I_2 + I_3$

Loop 1 - Start at "a"

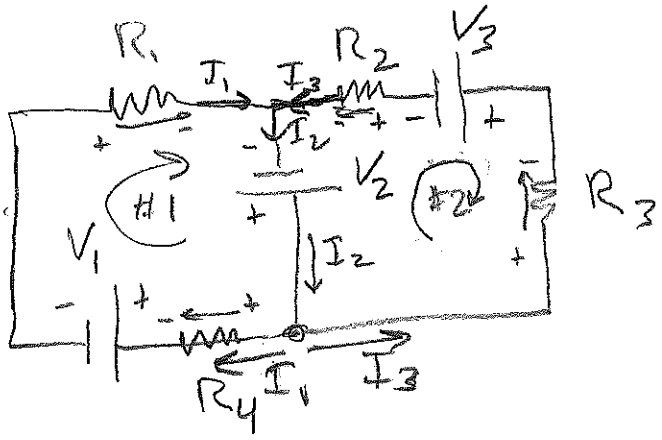
rise + drops = 0



$V_1 = 10$ V
$V_2 = 20$ V
$V_3 = 5$ V
$R_1 = 100\ \Omega$
$R_2 = 200\ \Omega$
$R_3 = 50\ \Omega$
$R_4 = 200\ \Omega$

See attached

21)



$V_1 = 10V$

$V_2 = 20V$

$V_3 = 5V$

$R_1 = 100\Omega$

$R_2 = 20\Omega$

$R_3 = 50\Omega$

$R_4 = 400\Omega$

What is the current in the circuit?

Junction rule

$I_2 = I_1 + I_3$

Loop rule

Loop #1

rise = drops

$V_2 = I_1 R_4 + V_1 + I_1 R_1$

$I_1(R_4 + R_1) = V_2 - V_1$

$I_1 = \frac{V_2 - V_1}{R_4 + R_1} = \underline{0.02\,A}$

Loop #2

rise = drop

$I_3 R_2 + V_3 + I_3 R_3 = V_2$

$I_3(R_2 + R_3) = V_2 - V_3$

$I_3 = \frac{V_2 - V_3}{R_2 + R_3}$

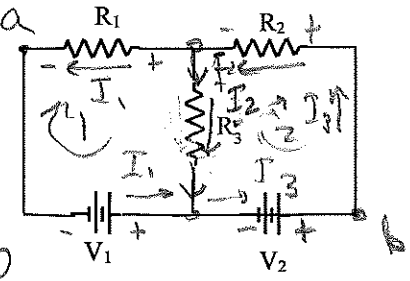
$I_3 = \underline{0.06\,A}$

$I_2 = I_1 + I_3$

$I_2 = \underline{0.08\,A}$

22) Use Kirchhoff's rules to solve for the current through each of the resistors R_1, R_2 and R_3.

$I_3 = I_1 + I_2$ — Junction rule



$V_1 = 12$ V
$V_2 = 9$ V
$V_3 = 5$ V
$R_1 = 7\ \Omega$
$R_2 = 8\ \Omega$
$R_3 = 4\ \Omega$

Start at 'a'

Loop 1

$$(+R_1 I_1) + (-R_3 I_2) + (-V_1) = 0$$

$$I_2 = \frac{R_1 I_1 - V_1}{R_3}$$

Loop 2 - start at b

$$(-V_2) + (+I_2 R_3) + (I_3 R_2) = 0$$

$$I_3 = \frac{V_2 - I_2 R_3}{R_2}$$

using Junction rule $\quad I_3 = I_1 + I_2$

$$(-V_2) + I_2 R_3 + (I_1 + I_2) R_2 = 0$$

Solve For I_2

$$I_2 (R_3 + R_2) = V_2 - I_1 R_2 \quad \Rightarrow \quad I_2 = \frac{V_2 - I_1 R_2}{(R_3 + R_2)}$$

Set equal to Loop 1

$$\frac{R_1 I_1 - V_1}{R_3} = \frac{V_2 - I_1 R_2}{R_3 + R_2}$$ Solve For I_1

$$I_1 R_1 (R_3 + R_2) - V_1 (R_3 + R_2) = V_2 R_3 - I_1 R_3 R_2$$

$$I_1 (R_1 (R_3 + R_2) + R_3 R_2) = V_1 (R_3 + R_2) + V_2 R_3$$

$$I_1 = \frac{V_1 (R_3 + R_2) + V_2 R_3}{(R_1 (R_3 + R_2) + (R_3 R_2))} = 1.55\ A$$

next page

$$I_2 = \frac{R_1 I_1 - V_1}{R_3} = -0.29$$

opposite direction

$$I_3 = \frac{V_2 - I_2 R_3}{R_2} = 1.27\text{A}$$

KEY

1) B
2) C
3) A
4) B
5) A) 200 Ω B) 0.06 A C) 0.06 A D) $V_1 = 1.2$ V, $V_2 = 2.4$ V, $V_3 = 3.6$ V $V_4 = 4.8$ V E) 0.72 W
6) A) 9.6 Ω B) 1.25 A C) $I_1 = 0.6$ A, $I_2 = 0.3$ A, $I_3 = 0.2$ A $I_4 = 0.15$ A D) V = 12 V for all resistors E) 15 W
7) 6 V
8) 0.5 A
9) B
10) C
11) C
12) A
13) 50 Ω
14) 0.8 A
15) 8 V
16) 8 V
17) 0.4 A
18) $I_1 = 0.32$ A, $I_2 = 0.24$ A $I_3 = 0.08$ A
19) $I_1 = 0.32$ A, $I_2 = 0.24$ A $I_3 = 0.08$ A
20) 0.009 A
21) $I_1 = 0.02$ A, $I_2 = 0.08$ A $I_3 = 0.06$ A
22) $I_1 = 1.55$ A, $I_2 = 0.29$ A $I_3 = 1.27$ A